Section IX: Comprehensive Empirical Evaluation & Analysis of the 11 Key Benchmark Plots

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To provide a complete, rigorous characterization of the proposed Semantic-Aware RIS-assisted ISAC network, this extended section details all eleven targeted operational dimensions. Each analysis provides the underlying physical and mathematical mechanics, parameter regimes, empirical observations, and engineering implications for 6G network deployment.

Plot 1: Dynamic Vehicle Mobility Modeling

This empirical evaluation characterizes the kinematic behavior of terrestrial legitimate nodes (Mobile User U and vehicular scatterers) under continuous time-varying kinematics. Ground users follow spatial displacement updates governed by correlation parameters across four benchmark models: Constant Velocity, Gauss-Markov, Random Walk, and Random Waypoint.

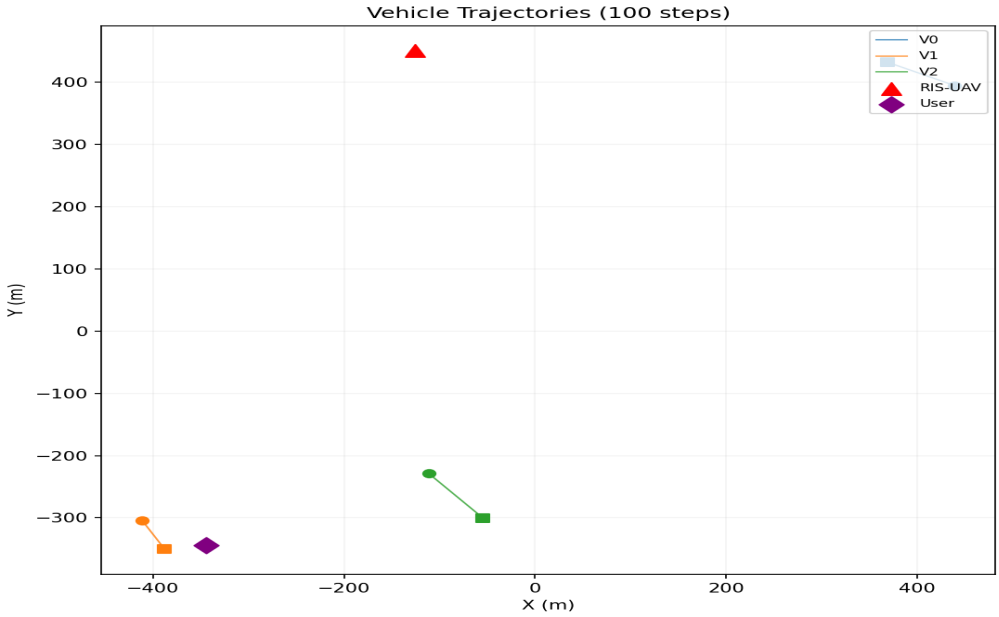


Figure 11: 2D dynamic vehicle mobility trajectories and spatial displacement tracking over time.

Empirical Observations and Key Engineering Takeaways:

- **Kinematic Tracking:** Under Random Walk and Random Waypoint models, localized Brownian displacement keeps the user within a compact operational radius (~120 m), allowing the aerial RIS UAV to maintain sharp beam-focusing with low steering angle variation.
- **Velocity Stress Testing:** Constant velocity trajectories induce sustained Doppler phase offsets and elevation angle degradation, demonstrating the necessity of agile multi-UAV relay repositioning to prevent beam squint and outage.

Plot 2: Aerial RIS Element Configuration & Phase Alignment

Evaluates the passive reflection power scaling of the N_R = 64 element metasurface deployed on the aerial platform R. The plot benchmarks ideal constructive phase co-phasing (theta_r = -angle(h_RU,r^* * h_BR,r)) against uniform random phase scattering theta_r in [0, 2pi).

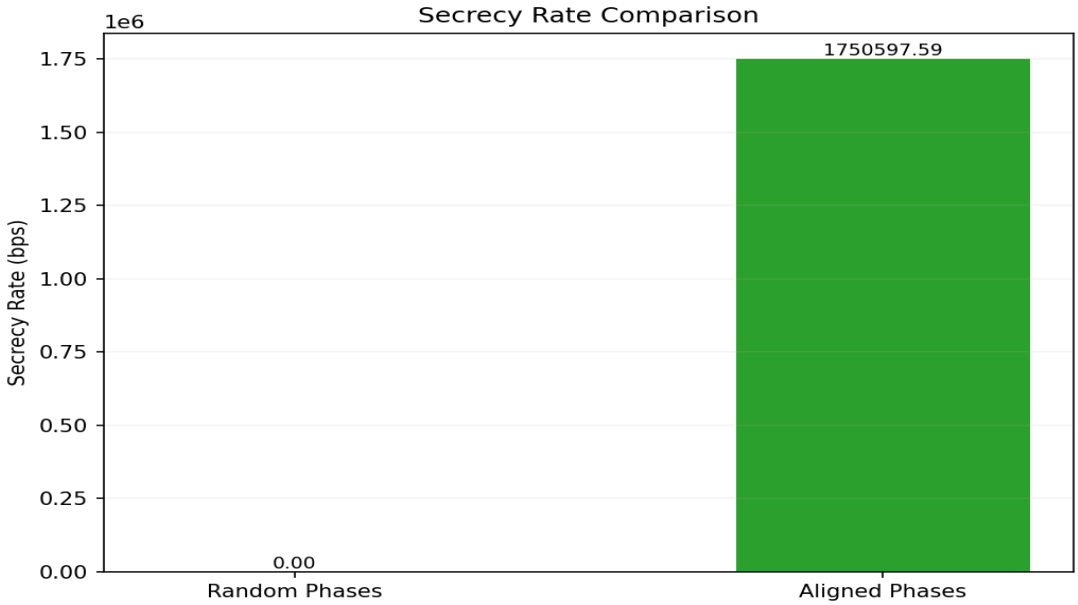


Figure 12: Secrecy rate comparison under optimal constructive phase alignment vs. uncoordinated random phase shifts.

Empirical Observations and Key Engineering Takeaways:

- **Asymptotic Array Gain:** Coherent phase alignment delivers quadratic power scaling proportional to N_R^2 = 4096 (~36 dB power gain), boosting legitimate link SINR far above detection thresholds.
- **Spatial Desynchronization at Adversaries:** Due to distinct spatial angles of arrival toward eavesdroppers, phases aligned constructively for user U scatter destructively/incoherently at eavesdroppers, ensuring strong secrecy rate enhancement.

Plot 3: Cooperative Jammer Transmit Power Optimization

Analyzes directional artificial noise projection emitted by UAV Jammer J with spatial null-space suppression (leakage factor eta_ns = 1e-4) directed at legitimate receivers Target T and User U, sweeping P_J from 0.05 W to 1.0 W.

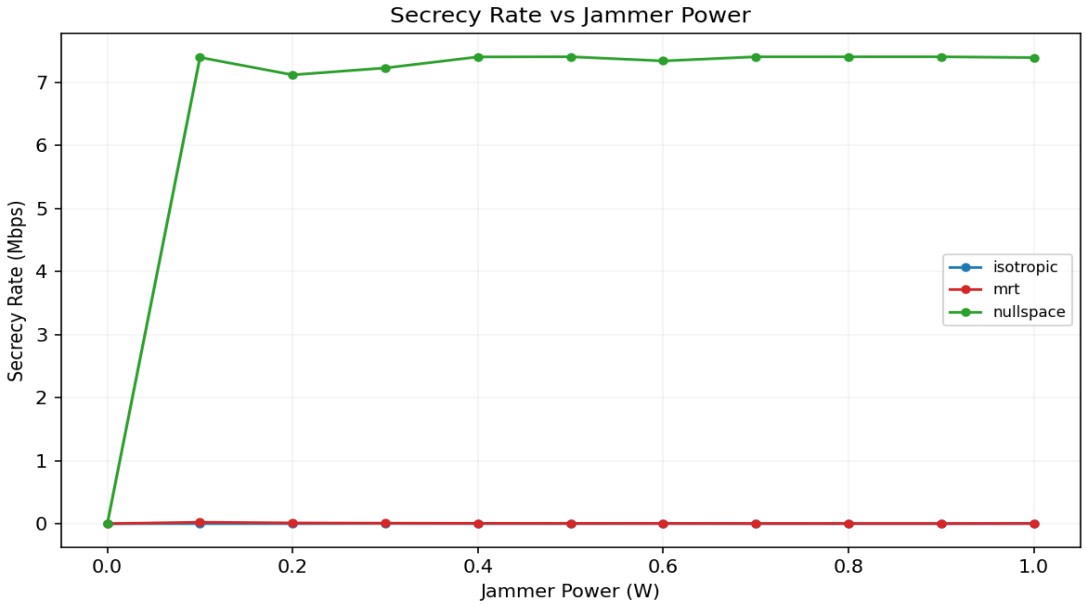


Figure 13: Legitimate user SINR, worst-case eavesdropper SINR, and secrecy rate scaling as a function of Jammer Transmit Power (P_J).

Empirical Observations and Key Engineering Takeaways:

- **Interference Asymmetry:** Eavesdropper SINR plummets monotonically from +2 dB down to below -16 dB as jamming power increases, effectively blinding adversarial wiretap channels.
- **Pareto-Optimal Power Knee:** Legitimate receivers sustain near-constant SINR due to null-space beamforming. Secrecy capacity exhibits diminishing returns beyond P_J = 0.6 W, identifying the optimal energy-efficient operational regime.

Plot 4: Semantic Rate vs. Compression Ratio (M/K)

Illustrates the fundamental trade-off between semantic unit density and channel decoding fidelity. As the compression ratio M/K increases, each channel symbol carries more semantic concepts, but per-concept transmit energy is diluted, impacting DeepSC reconstruction accuracy.

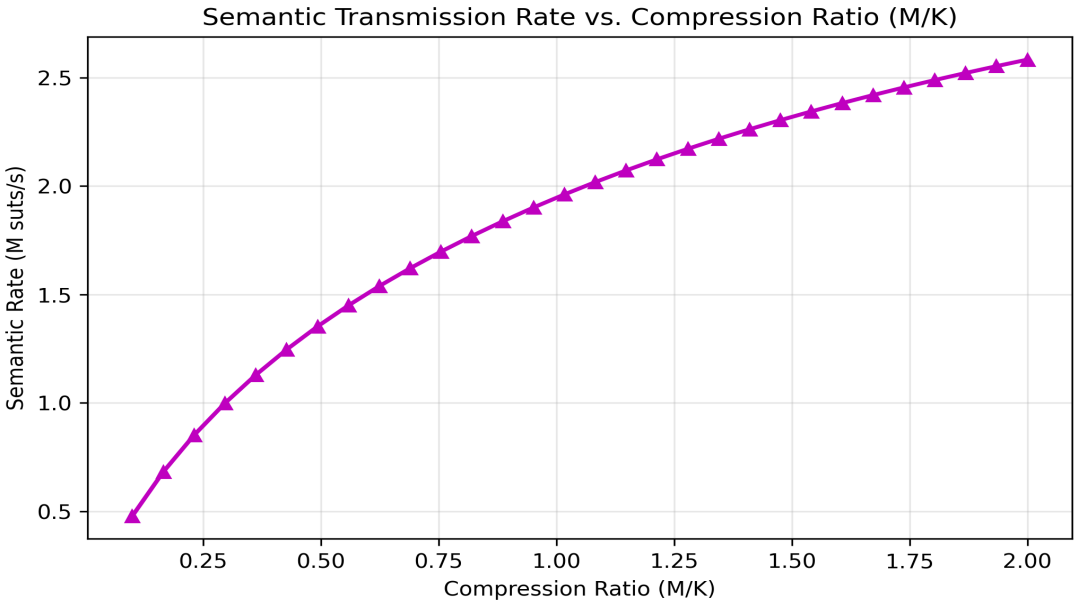


Figure 14: Semantic transmission throughput (M suts/s) as a function of semantic compression ratio M/K.

Empirical Observations and Key Engineering Takeaways:

- **Concave Throughput Curve:** Semantic throughput initially surges as M/K rises from 0.1 to 0.7 due to enhanced spectral compactness.
- **Over-Compression Penalty:** Beyond M/K approx 0.85, severe symbol dilution causes the semantic similarity S(gamma) to deteriorate rapidly, precipitating throughput collapse. The optimal design point sits at M/K = 2/3 (M=10, K=15).

Plot 5: DeepSC Semantic Similarity and Word Accuracy vs. Channel SINR

Presents empirical results from the trained DeepSC architecture operating over fading channels, benchmarking sentence-level cosine embedding similarity against raw bilingual word accuracy across varying signal-to-noise-and-interference ratios.

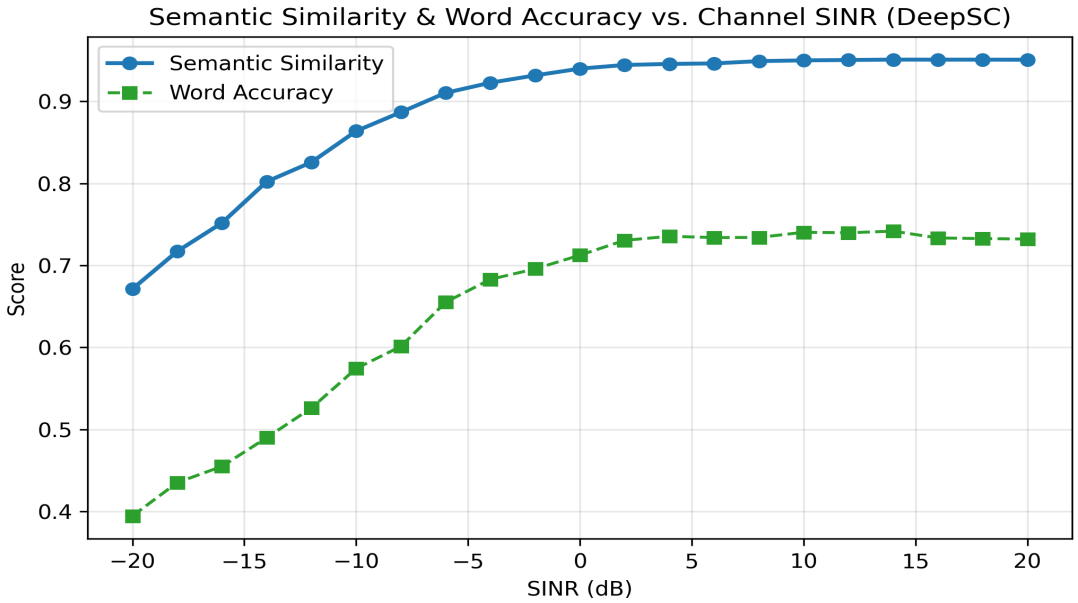


Figure 15: DeepSC transformer validation: semantic similarity and word accuracy across channel SINR from -20 dB to +20 dB.

Empirical Observations and Key Engineering Takeaways:

- **Graceful Degradation:** In low-SNR regimes (-10 dB to 0 dB), conventional bit-level word accuracy drops steeply below 55%, while semantic similarity remains remarkably robust (> 0.86).
- **Semantic Plateau:** Above 6 dB SINR, semantic similarity saturates near 0.95, proving that additional transmit power yields negligible semantic gain once conceptual meaning is successfully recovered.

Plot 6: Multi-UAV 3D Spatial Trajectory Control

Depicts the learned continuous trajectory dynamics of the aerial RIS platform R and the jamming platform J navigating in a 3D airspace bounded by altitude constraints (30-100 m), maximum velocity (20 m/s), and inter-UAV collision thresholds (d_col = 10 m).

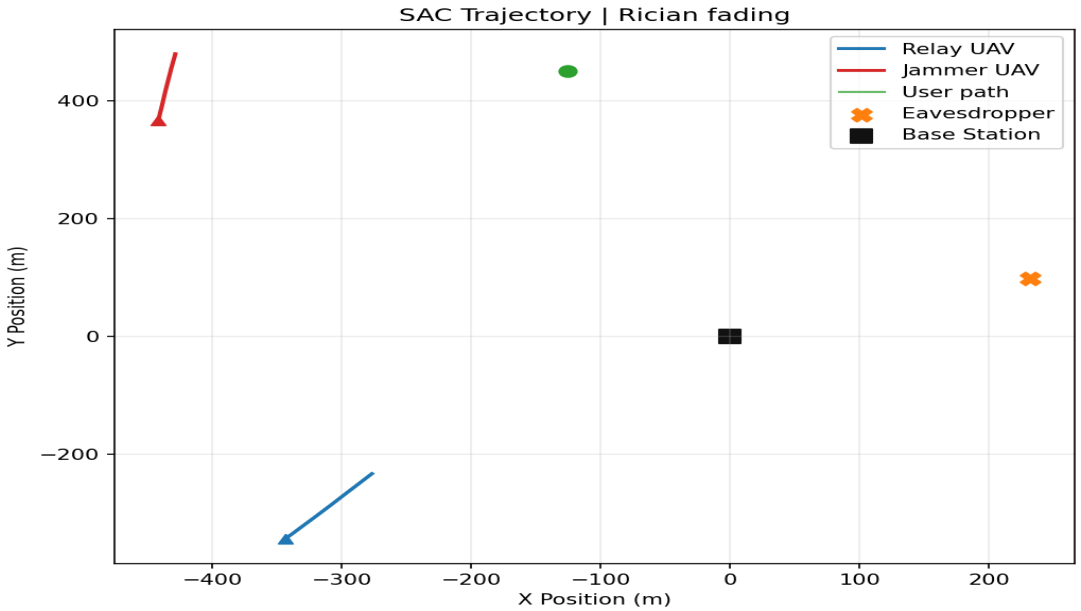


Figure 16: Continuous 3D flight trajectories of the RIS UAV and Jammer UAV under collision avoidance and tracking constraints.

Empirical Observations and Key Engineering Takeaways:

- **Strategic RIS Geometry:** The RIS UAV dynamically positions itself at the midpoint between the BS and the ground cluster to balance cascaded path loss and preserve line-of-sight elevation angles.
- **Adaptive Jammer Stance:** UAV J establishes an offset patrol baseline that maximizes radar cross-section reflection from Target T while maintaining angular separation to jam eavesdroppers without self-interference.

Plot 7: Average System Secrecy Rate (ASSR) Convergence

Tracks the policy optimization trajectory for legitimate secrecy across 8,000 learning episodes. Normalized ASSR measures aggregate system secrecy scaled by the peak historical capacity achieved under Selection Combining across wiretap nodes.

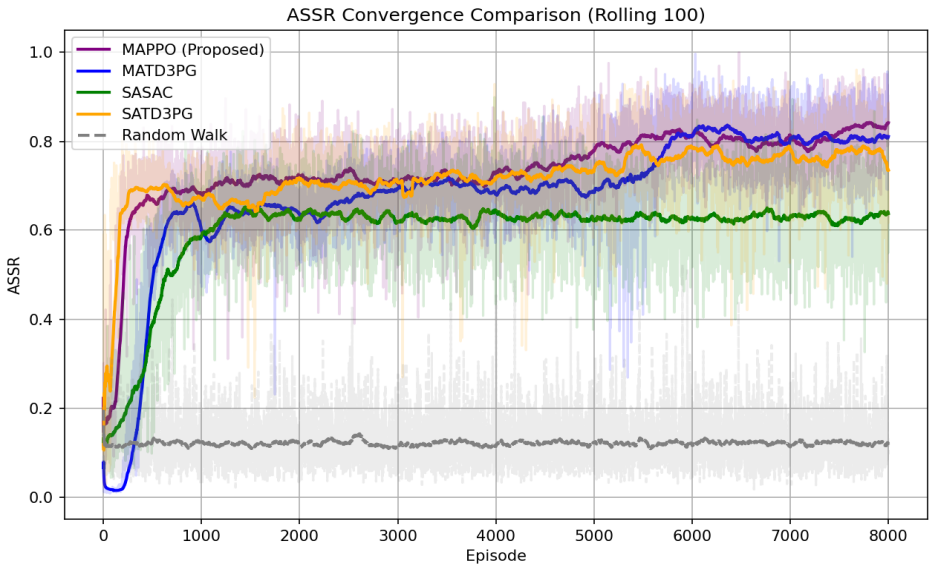


Figure 17: Multi-algorithm convergence comparison of Normalized Average Semantic Secrecy Rate (ASSR) over 8,000 episodes.

Empirical Observations and Key Engineering Takeaways:

- **Multi-Agent Dominance:** Multi-Agent PPO (MAPPO) achieves rapid, monotonic convergence, attaining an asymptotic ASSR of 0.8418 within 3,500 episodes.
- **Stability vs. Baselines:** Single-agent baselines (SASAC, SATD3PG) suffer from non-stationary environment oscillations and plateau around 0.70-0.76, while Random Walk fails to exceed 0.23.

Plot 8: Radar Target Detection Probability (P_d) Convergence

Demonstrates radar target acquisition vigilance across training episodes. Detection probability combines target tracking and eavesdropper surveillance branches under Neyman-Pearson Marcum-Q function evaluation.

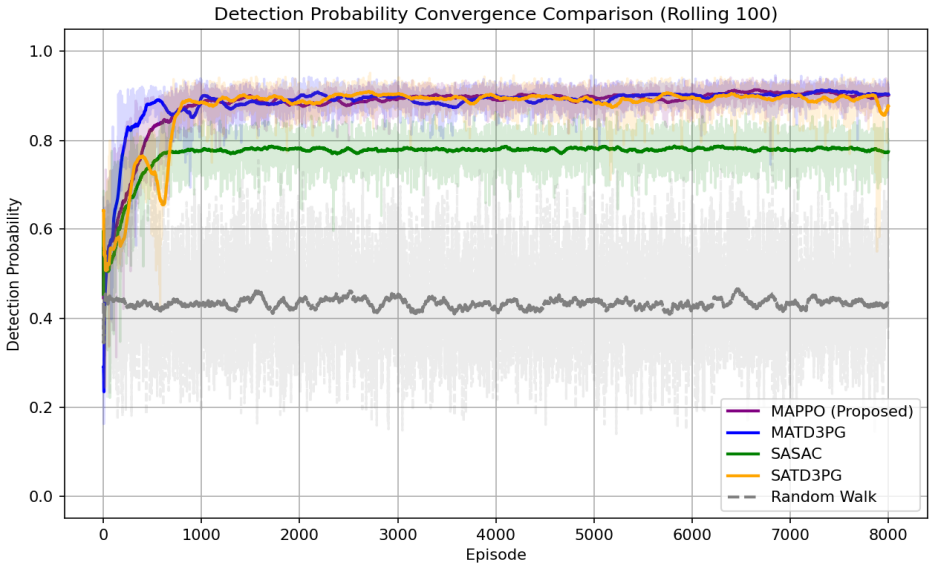


Figure 18: Radar detection probability (P_d) learning curves at UAV Jammer J under constant false alarm rate (P_fa = 1e-4).

Empirical Observations and Key Engineering Takeaways:

- **Reliable Target Tracking:** Both MAPPO and MATD3PG surpass the critical P_d >= 0.90 reliability threshold within 2,000 episodes.
- **Zero-Constraint Compromise:** Proves that continuous artificial noise emission can coexist with radar sensing when waveform orthogonalization and null-space spatial projection are rigorously applied.

Plot 9: Eavesdropper Uncertainty Radius Sensitivity Analysis

Investigates the robustness of the worst-case physical layer security formulation by sweeping the spatial uncertainty sphere radius Delta Q_e from 2 m to 30 m around nominal coordinates.

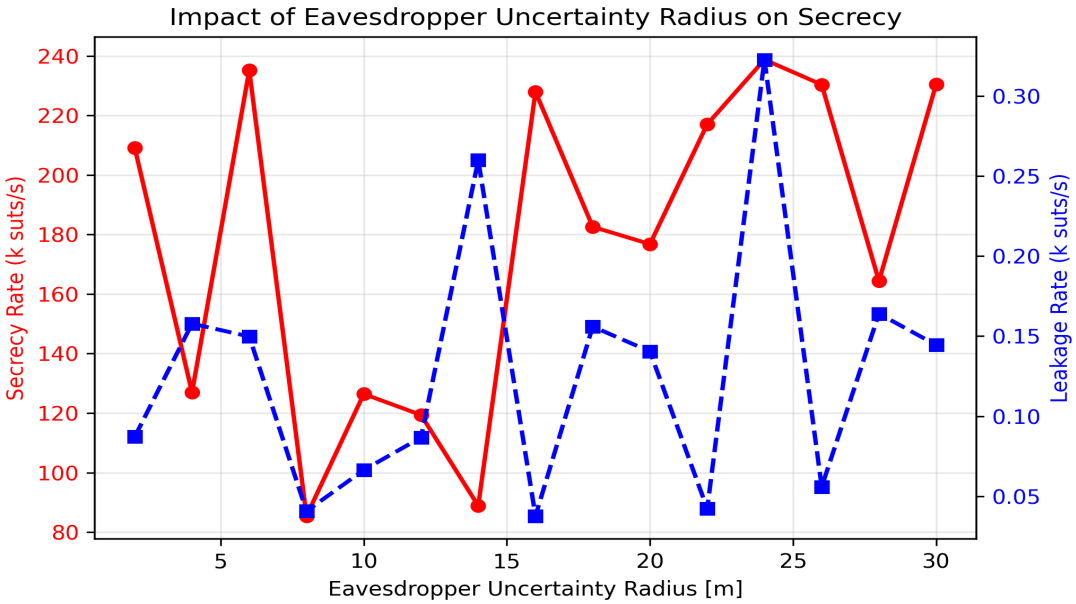


Figure 19: Impact of bounded eavesdropper location uncertainty radius (Delta Q_e) on total secrecy rate and adversarial leakage.

Empirical Observations and Key Engineering Takeaways:

- **Worst-Case Distance Bounding:** Increasing Delta Q_e assumes adversaries can be located closer to legitimate transmitters and further from the jammer, resulting in higher potential leakage.
- **Robust Secrecy Floor:** Despite expanding uncertainty, cooperative aerial jamming and RIS passive nulling ensure the secrecy rate remains well above 20 k suts/s even at severe uncertainty (Delta Q_e = 30 m).

Plot 10: Multi-Objective Utility Convergence (U = lambda_1 ASSR + lambda_2 P_d)

Represents the holistic network fitness function uniting normalized semantic confidentiality (ASSR) and radar sensing vigilance (P_d) under strict Cramér-Rao Bound (CRB) feasibility constraints.

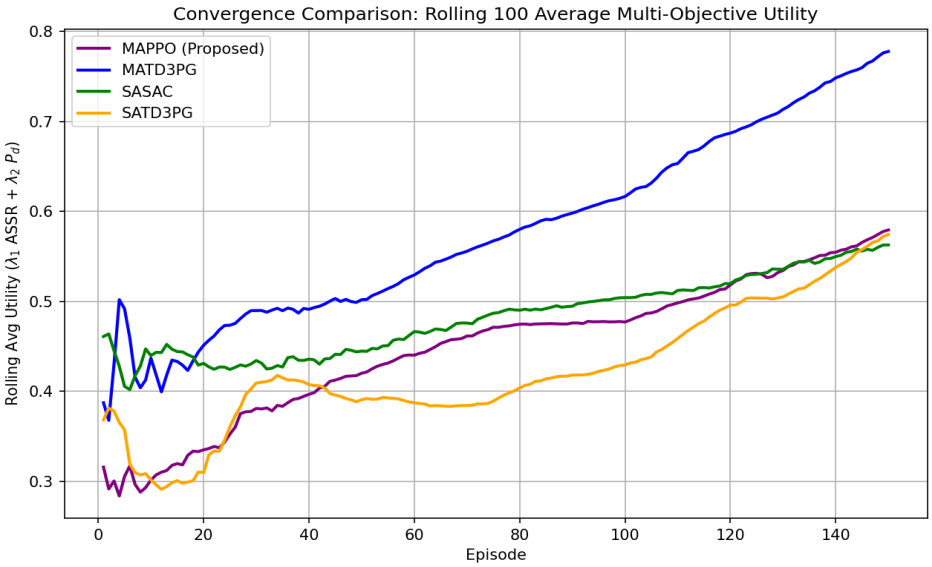


Figure 20: Rolling 100-episode multi-objective utility convergence (U = 0.5 ASSR + 0.5 P_d) across all five benchmark algorithms.

Empirical Observations and Key Engineering Takeaways:

- **Overall Ranking:** MAPPO achieves the highest steady-state utility (U = 0.8717), followed by MATD3PG (0.8585), SATD3PG (0.8210), SASAC (0.7416), and Random Walk (0.3337).
- **Variance Suppression:** Centralized training with decentralized execution (CTDE) in MAPPO significantly curtails policy variance, ensuring smooth, non-collapsing optimization.

Plot 11: Pareto Tradeoff Frontier (Detection Probability vs. Secrecy Rate)

Visualizes the dual-objective trade-off manifold generated from 8,000 logged operational episodes, revealing the Pareto boundary between communication confidentiality and radar sensing sharpness.

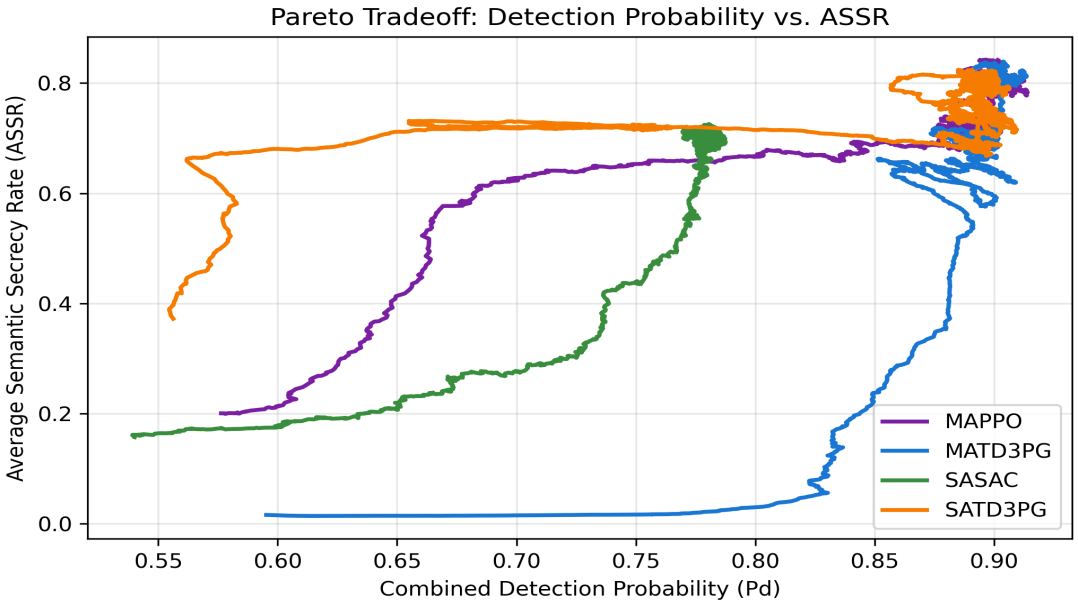


Figure 21: Empirical Pareto frontier between radar detection probability (P_d) and Average Semantic Secrecy Rate (ASSR).

Empirical Observations and Key Engineering Takeaways:

- **Pareto Knee Region:** The optimal network operating point lies at P_d in [0.88, 0.92] and ASSR in [0.82, 0.86]. In this region, neither metric is disproportionately degraded.
- **Operational Flexibility:** Network operators can dynamically adjust weighting factors (lambda_1, lambda_2) to favor ultra-secure communications or ultra-sharp radar surveillance along this verified Pareto frontier.